

DAC

ANUSHA N

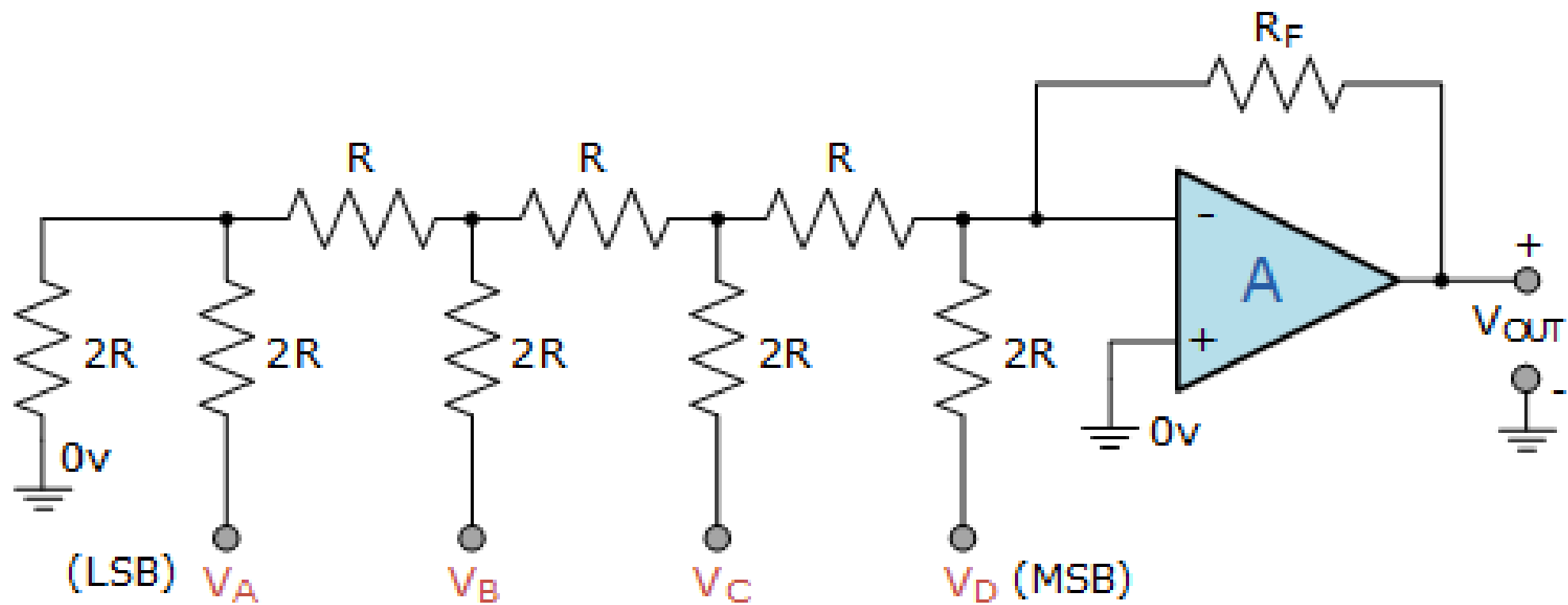
Types of ADC and DAC

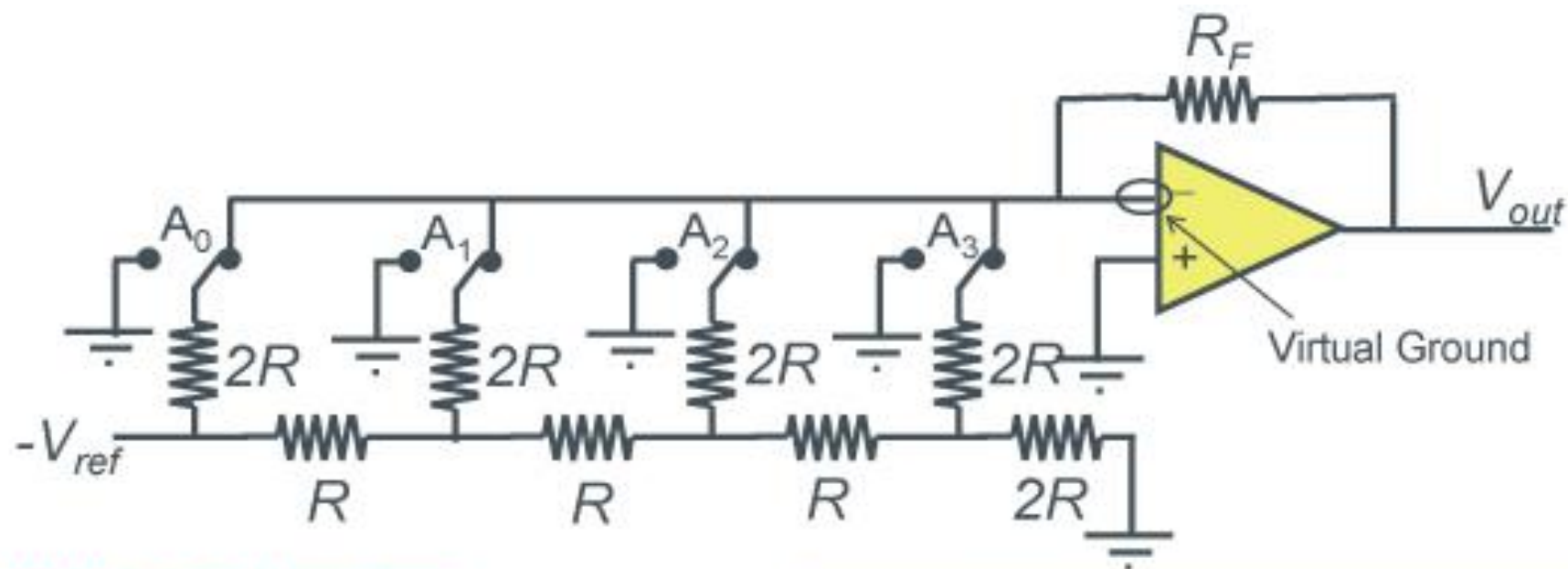
Types of ADC

- Flash ADC
- Successive Approximation ADC
- Single slope ADC

Types of DAC

- Voltage divider
- R-2R





Switch $\rightarrow$ Partial Resistance

Switch dimensions are made to be proportional to the amount of current

$MOSFET \left(\frac{W}{L} \right)$ is made proportional to the current

$$V_{out} = R_F \left(I_r \cdot A_0 + \frac{I_r}{2} \cdot A_1 + \frac{I_r}{2^2} \cdot A_2 + \frac{I_r}{2^3} \cdot A_3 \right)$$

$$I_r = \frac{V_{ref}}{2R}$$

$$V_0 = V_{\text{ref}} \left[\frac{D_0}{2} + \frac{D_1}{4} + \frac{D_2}{8} + \frac{D_3}{16} + \frac{D_4}{32} + \frac{D_5}{64} + \frac{D_6}{128} + \frac{D_7}{256} \right]$$

Ex:

1. If data = 00H [00000000], $V_{\text{ref}} = 10\text{V}$

$$V_0 = 10 \left[\frac{0}{2} + \frac{0}{4} + \frac{0}{8} + \frac{0}{16} + \frac{0}{32} + \frac{0}{64} + \frac{0}{128} + \frac{0}{256} \right]$$

Therefore, $V_0 = 0$

Volts.

2. If data is 80H [10000000], $V_{\text{ref}} = 10\text{V}$

$$V_0 = 10 \left[\frac{1}{2} + \frac{0}{4} + \frac{0}{8} + \frac{0}{16} + \frac{0}{32} + \frac{0}{64} + \frac{0}{128} + \frac{0}{256} \right]$$

Therefore, $V_0 = 5$

8051 Interfacing DAC

- Microcontroller are used in wide variety of applications like for measuring and control of physical quantity like temperature, pressure, speed, distance, etc.
- In these systems microcontroller generates output which is in digital form but the controlling system requires analog signal as they don't accept digital data thus making it necessary to use DAC which converts digital data into equivalent analog voltage
- Digital to Analog Converter is a device used to convert digital pulses to analog signals.
- In the figure shown, we use 8-bit DAC 0808. This IC converts digital data into equivalent analog Current. Hence we require an I to V converter to convert this current into equivalent voltage.
- DAC0808 provides 256 discrete voltage (or current) levels of output.

MC1408 DAC (OR DAC0808)(DAC CONVERTER INTERFACING)

DAC device use for convert digital pulse to analog signal .

Two methods of creating DAC ; binary weighted and R/2R LADDI

No of data bit input decide the resolution of DAC since no. of analog output level is equal to 2^n and n is the no. of data bit input .

8 input DAC provide 256 discrete voltage.

MC1408 the digital input converted to current (I_{out}) by connecting resistor of the I_{OUT} PIN

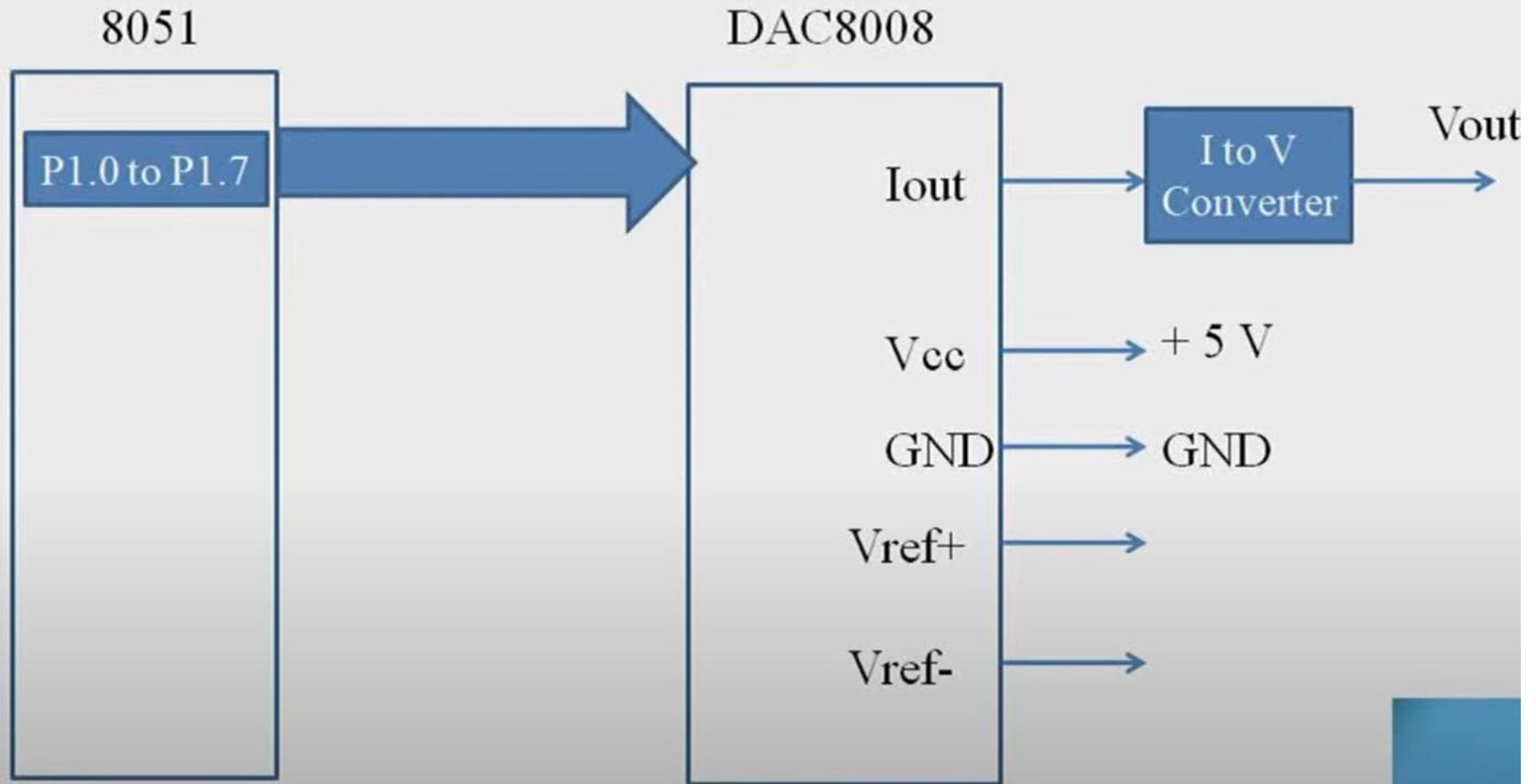
$$I_{out} = I_{ref} \left(\frac{D7}{2} + \frac{D6}{4} + \frac{D5}{8} + \frac{D4}{16} + \frac{D3}{32} + \frac{D2}{64} + \frac{D1}{128} + \frac{D0}{256} \right)$$

DAC 0808/09 signals

- D0-D7: Eight digital data input pins
- Iout: DAC Current output
- Vref+ and Vref-
- Vcc and GND
- CLK

DAC interfacing

-



SQUARE WAVE GENERATOR

```
BACK : MOV P1,# 0FFH  
      ACALL DELAY  
      MOV P1,#00H  
      ACALL DELAY  
      SJMP BACK
```

```
DELAY: MOV R7,#0FFH  
      L1:DJNZ R7,L1
```

Program for Sine Wave Generation

```
BACK:MOV DPTR, # SINE
      MOV R7,#18
L1:   MOV A,#00H
      MOVC A,@A+DPTR
      MOV P1,A
      INC DPTR
      DJNZ R7,L1
      SJMP BACK
```

```
SINE: DB  5,7,8,9,10,10,9,8,7,5,3,2,1,0,0,1,2,3
```

5+5*Sin (theta)

5

7

8

9

10

10

9

8

7

5

3

2

1

0

0

1

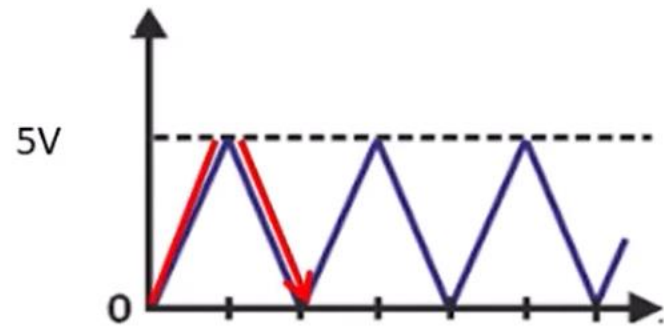
2

3

5 (360/0)

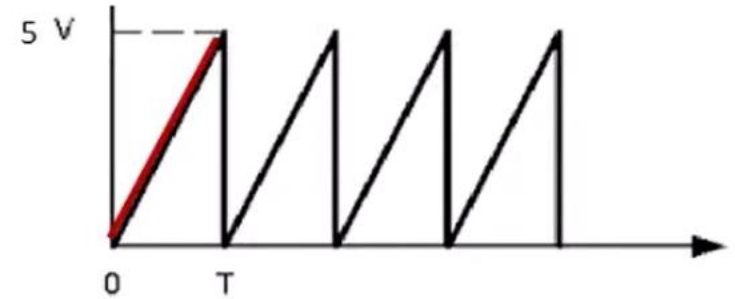
Program to Generate Triangular Signal

```
BACK:MOV A,#00H  
      L1: MOV P1,A  
          INC A  
      CJNE A,#0FFH,L1  
      L2: MOV P1,A  
          DEC A  
      CJNE A,#00H,L2
```



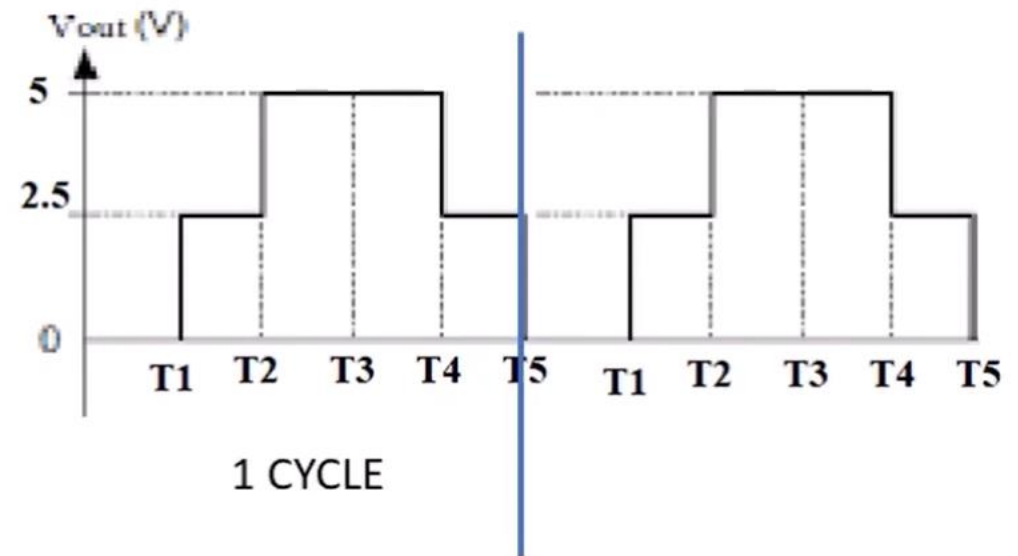
Program to Generate RAM/SAW Tooth Signal

```
BACK :  MOV A,#00H
        L1: MOV P1,A
          INC A
          CJNE A,#0FFH,L1
          MOV P1,A
          SJMP BACK
```



Program to Generate Staircase Signal

```
MOV P1,#00H  
ACALL DELAY  
MOV P1,# 7FH  
ACALL DELAY  
MOV P1,#0FFH  
ACALL DELAY  
ACALL DELAY  
MOV P1,#7FH  
ACALL DELAY
```



DAC program

- *Write assembly program to generate a square wave using DAC*

- *Algorithm*

1. Send 00h to port 1
2. Call a delay to get required off time
3. Send FFh to port 1
4. Call a delay to get required on time
5. Repeat steps 1 to 4 continuously

Program

```
AGAIN :MOV A, #00H  
MOV P1, A  
CALL DELAY  
MOV A, #FFH  
MOV P1, A  
CALL DELAY  
SJMP AGAIN
```

DAC program

- *Write assembly program to generate a triangular wave using DAC*

- *Algorithm*

1. Send 00h to FFh to port 1
2. Send FFh to 00h to port 1
3. Repeat steps 1 to 2 continuously

Think to write a delay subroutine

Program

```
AGAIN:MOV A, #00H  
RISE:MOV P1, A  
INR A  
CJNE A, #0FFH, RISE  
FALL:MOV P1, A  
DEC A  
CJNE A, #00H, FALL  
SJMP AGAIN
```

Digital-to-Analog (DAC)

General DAC Characteristics:

There are six key parameters you should consider when choosing a DAC.

A. Reference Voltage B. Resolution C. Linearity D. Speed E. Settling Time F. Error

A. Reference Voltage

To a large extent, the characteristics of a DAC are defined by its reference voltage.

1. Non-multiplier DAC: V_{Ref} is fixed (specified by the manufacturer)
2. Multiplier DAC: V_{Ref} is provided via an external source

B. Resolution

Resolution is a measure of precision, not accuracy. It is defined as the voltage change corresponding to changing the LSB.

1. Many options in the 8-16 bit range, with 12 bits being a typical cost / resolution trade off.
2. More bits More steps Greater Resolution

$$\text{Resolution} = V_{\text{LSB}} = V_{\text{Ref}} / 2^N$$

- Where N is the number of bits

C. Linearity

Ideally, a DAC will produce a linear relationship between a binary word and analog output

D. Speed

Usually specified as conversion or sampling rate.

1. High speed DACs are typically defined as $>1\text{MS/s}$ (1 MHz)
2. Some current 12-16 bit DACs can reach the 1GHz range
3. Conversion of input signal is limited by
 - Clock speed of the input signal
 - Settling time of the DAC

E. Settling Time

Ideally, an immediate change in analog voltage would occur when a new binary word enters into a DAC.

1. Settling time is the time taken by the DAC to reach $\frac{1}{2}$ of the LSB of its new voltage.
2. Components include delay, slew time, and ring time.
3. Fast converters reduce slew time, but usually result in longer ring times.
4. Delay time is normally a small term.

F. Errors

Because we do not live in an “ideal” world, considerations for possible error should be made.

1. **Non-Linearity:** *Occurs when analog output of signal is non-linear*
 - *Differential – analog step-sizes changes with increasing digital input (measure of largest deviation; between successive bits)*
 - **Integral** – *amount of deviation from a straight line after offset and gain errors removed; on concurrent bits*
1. **Non Monotonicity:** *Occurs when an increase in digital input results in a decrease in the analog output*
2. **Offset Error:** *Occurs when there is a constant offset between the actual output and the ideal Output*
3. **Gain Error:** *Occurs when the slope of the actual output deviates from the ideal output*

Resolution error: *Poor representations of ideal output due to poor resolution Size of voltage divisions affect the resolution*

Characteristics of DAC

- **Resolution**
- **Linearity**
- **Monotonicity**
- **Settling time**
- **Differential nonlinearity**

Characteristics of DAC

Digital-to-analog converters are normally chosen for application within a given system on the basis of their resolution and accuracy, which, in turn, is generally determined by N , the number of input bits of the converter. The following is a listing and definitions of key DAC terms and specifications:

Resolution

The resolution of a DAC is given by the number of bits, N . The resolution is the smallest increment of output that the DAC can produce. An 8-bit, DAC has a resolution of 8 bits, or one part in 2^8 . This results in a percentage of 0.39%.

Linearity

Linearity is the maximum allowable deviation from an ideal straight line drawn between the zero-scale and full-scale outputs. It is often given as a percentage or as a fraction of an LSB. For example, (12)LSB linearity for an 8-bit DAC corresponds to 0.195%. Some manufacturers call this parameter the nonlinearity.

Characteristics of DAC

Monotonicity

A DAC is said to be monotonic if each digital-code increment produces an output equal to or larger than that of the preceding code. A DAC is generally expected to be monotonic to increments as small as an LSB, but in general the smallest increment for which the DAC remains monotonic determines its monotonicity.

Settling time

The settling time is measured from the instant of a digital-input-code change to the time that the analog output reaches its corresponding new value to within a specified error band.

Differential nonlinearity

The differential nonlinearity of a DAC is the maximum deviation from ideal of the analog output between any two adjacent codes. It is expressed as a percentage of full scale or in terms of an LSB value.